

Exercise Sheet

Task 1 – Decimal to Binary Conversion

Convert the following decimal numbers into their binary representations. Make sure that you use enough bits to display the sign correctly.

a) $(125)_{10}$

b) $(39)_{10}$

Task 2 – Binary to Decimal Conversion

Convert the following unsigned binary numbers into their decimal representations by using the specified conversion method.

a) $(11101)_2$, conversion table

b) $(10101001)_2$, multiply and add

Task 3 – More Conversion

Convert the following numbers into their representation from the specified number system.

a) $(1011101)_2$ to the octal system

b) $(754)_8$ to the hexadecimal system

Task 4 – Two's Complement

Convert the following numbers into their two's complement representation. Use 8 bit to display the negative numbers.

a) $(-42)_{10}$

b) $(-15)_{10}$

Task 5 – Sign and Zero Extension

Use sign and zero extensions appropriately to put these 5-bit numbers into an 8-bit format.

- a) $(01010)_2$ signed
- b) $(11010)_2$ signed
- c) $(01010)_2$ unsigned
- d) $(11010)_2$ unsigned

Task 6 – Binary Operations

Compute the following operations.

- a) $(011010)_2 + (0101101)_2$
- b) $(1010)_2 * (1101)_2$
- c) $(110110)_2 XOR (101101)_2$
- d) $(1011010)_2 AND (1101110)_2$